URLs

HTTP is driven by URLs (Uniform Resource Locators), which are specially formed strings that contain enough information for a client to make the right request, for the right resource, at the right location. The client making the request will parse the URL and use the encoded information to drive the HTTP request. URLs are formed from up to five "components"…

- Scheme (usually the desired protocol)
- Authority (domain name, port number)
- Path
- Query (sometimes called "parameters")
- Fragment (sometimes called "anchor")

…and take the form `scheme://authority>path<?query><#fragment>`. For our purposes here, `<scheme>` will be `http` or `https`. The following examples illustrate this syntax; each URL conforms to the pattern of scheme, authority, path, and so on, though not all components are required for a URL to be valid.

- `http://some.server.com/insecure.php`
- `http://some.server.com/index.html#magic_id`
- `https://pault:notsosecret@my.domain.com:4430/chapters?name=How_the_internet_works`
- `file:///Users/pault/Documents/local_file.php`

First, we have `http://some.server.com/insecure.php`, which instructs an HTTP client to connect to the server at `some.server.com` on TCP 80 (the default port) and ask for the resource found at `/insecure.php` – all very straightforward (and unencrypted).

Next is a similar URL but with a fragment (or anchor) component, which acts like a "bookmark" within the resource: in the case of HTTP URLs, the browser will use this fragment data to scroll to wherever in the document a matching element identifier can be found. It's worth noting that the fragment data is never sent to the server; it is only used by the client.